



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

APPLIED CHEMISTRY

GENAC302

APPLY BASICS OF CHEMISTRY

Competence

RQF Level: 3

Learning Hours



30

Credits: 3

Sector: Building and construction services, Agriculture and food processing, Transport and logistics, Manufacturing and mining, Hospitality tourism, Art and craft, ICT and Multimedia

Trade: Building construction, Public works, Land surveying, Plumbing Technology, Interior design, Agriculture, Animal health, Wood Technology, Leather technology, Forestry, Water and Irrigation, Food processing, Automobile technology, Heavy machinery, Manufacturing technology, Mining technology, Food and beverage operations, Fine and plastic arts, Computer Systems and Architecture.

Module Type: General

Curriculum: TVET Certificate 3 in Building construction, Public works, Land surveying, Plumbing Technology, Interior design, Agriculture, Animal health, Wood Technology, Leather technology, Forestry, Water and Irrigation, Food processing, Automobile technology, Heavy machinery, Manufacturing technology, Mining technology, Food and beverage operations, Fine and plastic arts, Computer System and Architecture.

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Issue Date: April 2022

Purpose statement	This module describes the skills, knowledge and attitudes required to apply the basics of Chemistry. At the end of this module, the learner will be able to describe the structure of atom and molecules, apply the basics of inorganic chemistry and perform acid-base titration.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	100%
	Practical work:		70%		70%	
	● Group project and presentation	20%				
	● Individual project /Work	50%				
			Summative Assessment		0%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Describe the structure of atom and molecules	1.1. An atom is properly explained according to Dalton's atomic theory
	1.2. Atomic mass is correctly determined based on periodic table of chemical elements
	1.3. Chemical molecule formation is appropriately explained in accordance with chemical properties of elements
	1.4. Transition metals are properly discussed according to their physical and chemical properties
2. Apply the basics of inorganic chemistry	2.1. Chemical equations are correctly balanced based on types of chemical reactions
	2.2. Acids, Bases and Amphoteric substances are properly distinguished as per their chemical properties and uses
	2.3. Concentration of chemical solution is accurately calculated based on expressions of concentration
3. Perform acid-base titration	3.1. Laboratory apparatus and reagents are properly identified according to acid-base titration
	3.2. Acid-base titration is correctly conducted according to the concentration of solution

	3.3. Utility of acid-base titration is clearly explained refer to its area of application
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Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Describe the structure of atom and molecules 2. Apply the basics of inorganic chemistry 3. Perform acid-base titration
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Learning outcome 1: Describe the structure of atom and molecules	Learning hours: 10
Indicative content	
● Description of atom ✓ Dalton's atomic theory ✓ Atomic particles structure and their properties ● Nucleus ● Electrons ● Calculation of Atomic mass ✓ Periodic table of chemical elements ✓ Atomic mass, Relative atomic mass and Isotopes ✓ Electronic configuration ● Molecule formation ✓ Properties of metals and non-metals ✓ Chemical bonds ● Explanation of transition metals ✓ Physical and chemical properties ✓ Uses of transition metals	
Resources required for the learning outcome	
Equipment	- White board, black board, Projector, Computer
Materials	- Chalk, Marker pen, Flip Chart, internet bundles

Tools	- Reference books, Periodic table of chemical elements, Calculator
Facilitation techniques	- Demonstration and simulation, Individual and group work
Formative assessment methods	- Written assessment, Oral presentation, Performance assessment

Learning outcome 2: Apply the basics of inorganic chemistry	Learning hours: 10
Indicative content	
● Balancing chemical equations ✓ Types of chemical reaction ✓ Balance of chemical equations ● Distinction of Acids, Bases and Amphoteric substances ✓ Chemical properties of Acids, Bases and Amphoteric substances ✓ Use of acids and bases ● Calculation of molar concentration ✓ Mole concept ● Number of moles ● Relative molecular mass (RMM) ✓ Types of solution ✓ Calculation of Molarity (C), Normality (N), Molality (M), Percentage composition (%).	
Resources required for the indicative content	
Equipment	- White board, black board, Projector, Computer
Materials	- Chalk, Marker pen, Flip Chart, internet bundles, solute, solvent
Tools	- Reference books, Periodic table of chemical elements, Calculator, dropper, flask, beaker, test tube, PH meter
Facilitation techniques	- Demonstration and simulation, Individual and group work, Laboratory experiment, Trainer guided
Formative assessment methods	- Written assessment, Oral presentation, Performance assessment, Product based assessment

Learning outcome 3: Perform acid-base titration		Learning hours: 10	
Indicative content			
● Identification of laboratory apparatus and reagents for Acid-base titration ✓ Acid-base titration ✓ Apparatus and reagents for Acid-base titration ● Conduct acid-base titration ✓ Preparation of solutions ● Solute ● Solvent ● Solution and standard solution ✓ Dissolution and Dilution ✓ Results interpretation ● Application of acid-base titration ✓ Purpose of titration ✓ Areas of application of acid-base titration			
Resources required for the indicative content			
Equipment	- White board, black board, Projector, Computer		
Materials	- Chalk, Marker pen, Flip Chart, internet bundles, solute, solvent		
Tools	- Reference books, Periodic table of chemical elements, Calculator, dropper, flask, beaker, test tube,PH meter		
Facilitation techniques	- Demonstration and simulation, Individual and group work, Laboratory experiment, Trainer guided		
Formative assessment methods	- Written assessment, Oral presentation, Performance assessment, Product based assessment		

References:

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6. Hushes, P. C. (2002). Chemistry for advanced level. London: JOHN MURRAY.
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